

EXPLORATORY DATA ANALYSIS (EDA) & DATA CLEANING

Week 1 – Task 3

Dataset: Titanic Dataset

1. Introduction

Exploratory Data Analysis (EDA) and data cleaning are important steps in any data science project. Before performing analysis or building machine learning models, it is necessary to understand the structure and quality of the dataset.

For this task, the Titanic dataset was selected. The dataset contains information about passengers who travelled on the Titanic, including their age, gender, passenger class, fare, embarkation port, and survival status.

The main objective of this project was to inspect the dataset, identify data-quality problems, clean the data, perform exploratory analysis, visualize important patterns, and identify relationships between numerical variables.

The analysis was performed using Python, Pandas, NumPy, Matplotlib, and Seaborn.

2. Dataset Overview

The Titanic dataset contains information about individual passengers.

Dataset Features

Column	Description
PassengerId	Unique identification number of the passenger
Survived	Survival status (0 = Did not survive, 1 = Survived)
Pclass	Passenger class (1st, 2nd, or 3rd)
Name	Name of the passenger
Sex	Gender of the passenger
Age	Age of the passenger
SibSp	Number of siblings or spouses aboard
Parch	Number of parents or children aboard

Column	Description
Ticket	Ticket number
Fare	Passenger fare
Cabin	Cabin number
Embarked	Port from which the passenger embarked

The original dataset contains **891 rows and 12 columns**.

3. Tools and Technologies Used

The following Python libraries were used:

- **Pandas** – for data loading, manipulation, and cleaning
 - **NumPy** – for numerical operations
 - **Matplotlib** – for creating visualizations
 - **Seaborn** – for statistical visualization
 - **Jupyter Notebook** – for documenting and executing the analysis
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4. Data Inspection

The dataset was initially inspected using several Pandas functions.

The following operations were performed:

- Displayed the first five records using `head()`
- Displayed the last five records using `tail()`
- Checked the number of rows and columns using `shape`
- Examined column names
- Checked data types using `dtypes`
- Examined dataset structure using `info()`
- Generated statistical summaries using `describe()`

This initial inspection helped in understanding the structure and characteristics of the dataset before cleaning.

5. Missing Value Analysis

Missing values were checked for every column, and their percentages were calculated.

The main columns containing missing values were:

- **Age**
- **Cabin**
- **Embarked**

A missing-value heatmap was also created using Seaborn to visually identify missing data.

Missing Value Treatment

Age

The Age column contains missing numerical values. Instead of removing these records, the missing values were replaced using the **median age**.

The median was selected because it is less affected by extreme values than the mean.

Embarked

The Embarked column is categorical and contains a small number of missing values. These were replaced using the **mode**, which represents the most frequently occurring category.

Cabin

The Cabin column contains a large proportion of missing values. Directly filling these values with artificial cabin numbers would not provide meaningful information.

Therefore, a new feature called **Cabin_Known** was created.

- 1 indicates that cabin information was available.
- 0 indicates that cabin information was missing.

After creating this feature, the original Cabin column was removed.

6. Duplicate Data

Duplicate rows were checked using the Pandas duplicated() function.

Any duplicate records identified were removed using drop_duplicates().

After the cleaning process, the dataset was checked again to ensure that duplicate rows were no longer present.

7. Data Type Correction

The data types of the columns were examined during the cleaning process.

The following columns were converted to categorical data types:

- Survived
- Pclass

This is appropriate because these columns represent categories rather than continuous numerical measurements.

8. Outlier Detection

Outliers were investigated using box plots for numerical variables, particularly:

- Age
- Fare

The **Interquartile Range (IQR)** method was used to identify potential outliers.

IQR Formula

$$\text{IQR} = Q3 - Q1$$

The lower and upper boundaries were calculated using:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Bound} = Q3 + 1.5 \times \text{IQR}$$

Values outside these boundaries were considered potential outliers.

9. Outlier Treatment

The Fare column contained some extremely high values.

These values may represent genuine high-priced tickets rather than incorrect data. Therefore, the records were not deleted.

Instead, IQR-based capping was applied to the Fare column. Extreme values were limited to the calculated upper IQR boundary.

This approach reduces the influence of extreme values while preserving the passenger records.

10. Exploratory Data Analysis

Several visualizations were created to understand patterns in the Titanic dataset.

10.1 Survival Distribution

A count plot was created to compare passengers who survived with passengers who did not survive.

The visualization shows that the number of passengers who did not survive was greater than the number who survived.

10.2 Survival by Gender

A count plot was used to compare survival outcomes between male and female passengers.

The analysis shows a clear difference in survival outcomes between genders. Female passengers generally had a much higher survival rate than male passengers.

This indicates that gender was an important factor associated with survival.

10.3 Survival by Passenger Class

Survival was also analyzed according to passenger class.

Passengers in **1st class generally had better survival outcomes** compared with passengers in 2nd and 3rd class.

Passengers in 3rd class had comparatively lower survival rates.

This suggests that passenger class was an important factor associated with survival.

10.4 Age Distribution

A histogram was created to understand the distribution of passenger ages.

The majority of passengers were adults, while children represented a smaller portion of the dataset.

The distribution also contains passengers across a wide range of ages.

10.5 Fare Distribution

A histogram was created to examine passenger fares.

Most passengers paid relatively lower fares, while a smaller number of passengers paid considerably higher fares.

This explains the presence of high-value observations detected during outlier analysis.

10.6 Survival by Embarkation Port

Survival was also compared across the three embarkation ports.

The visualization showed differences in the number of survivors and non-survivors among passengers from different embarkation ports.

However, these differences may also be influenced by passenger class and other variables.

11. Correlation Analysis

A correlation matrix was generated for the numerical variables.

A Seaborn heatmap was used to visualize the correlation values.

Correlation values range from **-1 to +1**.

- A value close to **+1** indicates a strong positive relationship.
- A value close to **-1** indicates a strong negative relationship.
- A value close to **0** indicates a weak or no linear relationship.

Top Three Observations

1. Fare and Passenger Class

There is a noticeable relationship between passenger class and fare. Higher-class passengers generally paid higher fares.

2. Fare and Survival

Fare has a positive relationship with survival. Passengers who paid higher fares generally had better survival outcomes.

3. Passenger Class and Survival

Passenger class is associated with survival. Higher-class passengers generally had higher survival rates than lower-class passengers.

It is important to note that **correlation indicates association and does not prove causation**.

12. Challenges Faced

Several challenges were encountered during the data-cleaning and EDA process.

Missing Values

The Age, Cabin, and Embarked columns contained missing values. Different approaches had to be selected depending on the type and amount of missing data.

Cabin Data

The Cabin column contained a large number of missing values. Instead of using arbitrary values, a separate Cabin_Known feature was created to preserve useful information.

Outliers

The Fare column contained extreme values. It was necessary to determine whether these values represented errors or genuine observations. IQR capping was used rather than deleting potentially valid records.

Data Types

Some columns represented categorical information even though they were initially stored as numerical values. Appropriate data types were therefore assigned.

13. Final Data Cleaning Summary

The following cleaning operations were completed:

1. Loaded the Titanic dataset using Pandas.
2. Inspected the structure and statistical properties of the data.
3. Identified missing values and calculated their percentages.
4. Filled missing Age values using the median.
5. Filled missing Embarked values using the mode.
6. Created a Cabin_Known feature.
7. Removed the original Cabin column because of extensive missing data.
8. Identified and removed duplicate rows.
9. Converted Survived and Pclass to categorical data types.
10. Detected potential outliers using the IQR method.
11. Capped extreme Fare values using IQR boundaries.

12. Verified the final dataset for missing values and duplicates.

13. Exported the cleaned dataset as a new CSV file.

14. Conclusion

The Titanic dataset was successfully cleaned and analyzed using Python.

The project demonstrated the complete workflow of a basic data science exploratory analysis, starting from raw data inspection and continuing through data cleaning, outlier detection, visualization, and correlation analysis.

The analysis showed that factors such as **gender, passenger class, and fare were strongly associated with passenger survival.**

The project also demonstrated the importance of selecting appropriate methods for handling missing values and outliers rather than applying the same technique to every column.

The final cleaned dataset was exported separately so that the original raw dataset remained unchanged.

This task provided practical experience with Pandas, Matplotlib, Seaborn, data cleaning techniques, statistical analysis, and exploratory data visualization.